Exponential and Logarithmic Equations

Name _____

Algebra 2 Lesson 6-6 (fast track) Date _____ Period _____

Learning goal: I can solve an exponential equation by rewriting both sides with a common base, and solve a logarithmic equation by setting the arguments equal and checking for extraneous solutions.

Language goal: I can explain each step of my solution using the words *common base*, *exponent*, *argument* and *extraneous solution*.

Spanish: exponential equation = *ecuación exponencial*; logarithmic equation = *ecuación logarítmica*.

On the quiz: You will (1) solve an exponential equation with no common base by taking the log of both sides (full Savvas quiz only; optional IXL 9F2); (2) rewrite two clouds' exponential expressions, which have different bases, with a common base to find when they are equal, then find the volume (Example 1; the bonus); (3) find and fix the mistake in someone's common-base work and name each step (Example 1 and the Practice); (4) solve a log equation with a coefficient in front of one log (move it inside first) and keep only the solutions that make every argument positive (Example 5); (5) solve a log equation by graphing (full Savvas quiz only).

On the Topic 6 test: Question 3 asks you to rewrite an expression by putting x in place of $s - t$; no Example in this lesson teaches it, so the quiz is the goal.

Example 1: Solve Exponential Equations Using a Common Base

Property of Equality for Exponential Equations: if $b > 0$ and $b \neq 1$, then $b^x = b^y$ if and only if $x = y$. Two powers of the same base are equal exactly when their **exponents** are equal.

What is the solution to $\left(\frac{1}{2}\right)^{x+7} = 4^{3x}$? The key: $\frac{1}{2}$ and 4 are both powers of 2, so 2 is a **common base**.

Line	Work	Why
1	$\left(\frac{1}{2}\right)^{x+7} = 4^{3x}$	the equation
2	$(2^{-1})^{x+7} = (2^2)^{3x}$	rewrite each side with a common base: $\frac{1}{2} = 2^{-1}$, $4 = 2^2$
3	$2^{-x-7} = 2^{6x}$	Power of a Power Property: multiply the exponents
4	$-x - 7 = 6x$	Property of Equality for Exponential Equations
5	$-7 = 7x$	add x to both sides
6	$x = -1$	divide by 7

Check $x = -1$: $\left(\frac{1}{2}\right)^{-1+7} = \left(\frac{1}{2}\right)^6 = \frac{1}{64}$ and $4^{3(-1)} = 4^{-3} = \frac{1}{64}$. ✓

Watch line 3. $(2^{-1})^{x+7}$ multiplies -1 by the *whole* exponent $x + 7$: $-x - 7$, not $-x + 7$.

Say it: I wrote both sides with the common base _____. Then I set the exponents _____ equal because ...

Try It 1: Solve each equation using a common base

Hint: for each part, name the common base, rewrite both sides, set the exponents equal, solve, and check.

a. $25^{3x} = 125^{x+2}$

Common base: _____

$x =$ _____ Check: _____

b. $0.001 = 10^{6x}$

Common base: _____

$x =$ _____ Check: _____

Practice: Find the mistake

Lena solved $2^{x+5} = 8^{x-1}$, but she made an error. Her work is shown.

Line	Lena's work	Step
1	$8^{x-1} = 2^{x+5}$	
2	$(2^3)^{x-1} = 2^{x+5}$	_____
3	$2^{3x-1} = 2^{x+5}$	_____
4	$3x - 1 = x + 5$	_____
5	$x = 3$	

a. Name the step in lines 2, 3 and 4. Choose from:

- Rewrite with a common base
- Power of a Power Property
- Property of Equality for Exponential Equations

b. Lena made an error in line _____. The line should be _____. The correct solution is $x =$ _____.

c. Check your solution in $2^{x+5} = 8^{x-1}$:

Example 5: Solve Logarithmic Equations

Property of Equality for Logarithmic Equations: if $b > 0$, $b \neq 1$, and x and y are positive, then $\log_b x = \log_b y$ if and only if $x = y$: the **arguments** (the expressions inside the logs) are equal.

What is the solution to $\ln(x^2 - 16) = \ln(6x)$?

Work	**Why**
$\ln(x^2 - 16) = \ln(6x)$	the equation
$x^2 - 16 = 6x$	Property of Equality for Logarithmic Equations
$x^2 - 6x - 16 = 0$	subtract $6x$
$(x - 8)(x + 2) = 0$	factor
$x = 8$ or $x = -2$	Zero Product Property

Check each value in the *original* equation.

$x = 8$: $\ln(8^2 - 16) = \ln(6 \cdot 8)$ $\ln 48 = \ln 48$ ✓
$x = -2$: $\ln((-2)^2 - 16) = \ln(6 \cdot (-2))$ $\ln(-12) = \ln(-12)$ ×

A log with a coefficient? Move the coefficient inside first, $k \log x = \log x^k$ for $x > 0$ (**Power Property**), then set the arguments equal.

Logarithms are not defined for zero or negative arguments, so only $x = 8$ is a solution. The value $x = -2$ is an **extraneous solution**: it solves the quadratic $x^2 - 6x - 16 = 0$ but not the original log equation.

Say it: $x = -2$ is an extraneous solution because the argument $6x$ would be _____, and ...

Try It 5: Solve each equation

Hint: set the arguments equal, solve, and check every value in the original equation.

a. $\log_5(x^2 - 45) = \log_5(4x)$

Solution: _____ Extraneous: _____

b. $\ln(-4x - 1) = \ln(4x^2)$

Solution: _____ Extraneous: _____

Is a negative solution always extraneous? Check both *arguments*, not just x.

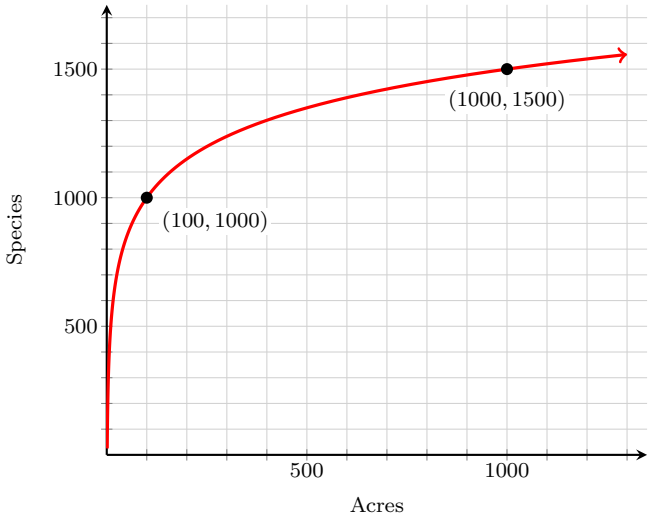
IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
YQY ★	Solve exponential equations by rewriting the base	Quiz 2 and 3 (Example 1)
BXU ★	Solve logarithmic equations I	Quiz 4 (Example 5) and Exit question b
RLX ★	Solve logarithmic equations II	Quiz 4: a log with a coefficient in front
V2J	Power rule	Warm-up for quiz 3: the Power of a Power line
7T3	Power property of logarithms	Warm-up for quiz 4: $k \log x = \log x^k$
9F2	Solve exponential equations using common logarithms	Optional: quiz 1 (full Savvas quiz only)

Exit question: Practice 43 (Higher Order Thinking)

A biologist is using the logarithmic model $n = k \log(A)$ to determine the number of a species n that can live on a land mass of area A. The constant k varies according to the species.



a. Use the graph to determine the constant k for the species that the scientist is studying.

$k =$ _____

b. Determine the land mass in acres that is needed to support 3,000 of the species.

$A =$ _____ acres

Before the quiz, finish these sentences.

To solve $25^{3x} = 125^{x+2}$, I first write both sides with the common base _____, and then I set the _____ equal.

To solve $\log_5(x^2 - 45) = \log_5(4x)$, I set the _____ equal, and I check each answer because a value that makes an argument zero or negative gives an _____.